LAB 3 — ARCHITECTURE SELECTION & JUSTIFICATION

Project: Student Project Portfolio & Showcase App
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Architecture Selection: Layered Architecture

We have selected **Layered Architecture** for the Student Project Portfolio & Showcase App. The system is organized into three layers: Presentation, Business Logic, and Data. The Presentation Layer provides the interface for students, judges, and public voters. The Business Logic Layer handles project submissions, GitHub repository validation, digital evaluation, voting security, authentication, and leaderboard calculations. The Data Layer stores and retrieves project details, user information, evaluations, and voting records.

Reason 1 — Separation of Responsibilities: Each layer has a clearly defined role. User interface changes can be managed separately from business rules and database operations. This makes the system easier to understand, test, and maintain.

Reason 2 — Simplicity and Maintainability: Layered Architecture provides a straightforward structure for the application's related functions without the additional deployment and communication complexity of Microservices. Individual business components can be organized according to their responsibilities.

Security Advantage: Authentication and Access Control can verify user identities and roles before allowing operations. The Voting and Anti-Sybil Security component can reject duplicate or suspicious voting attempts, helping protect the integrity of public voting results.

Performance Benefit: Keeping related business functionality within the same application can reduce the network overhead associated with communication between separately deployed services. Efficient database queries and indexing can further help voting and leaderboard operations meet the Lab 1 target of API response latency under 300 milliseconds. This target must be verified through performance testing.

Component Diagram — Student Project Portfolio & Showcase App

